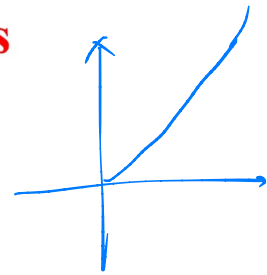


Stability Theorems for Nonautonomous Systems



Comparison Functions

Definition

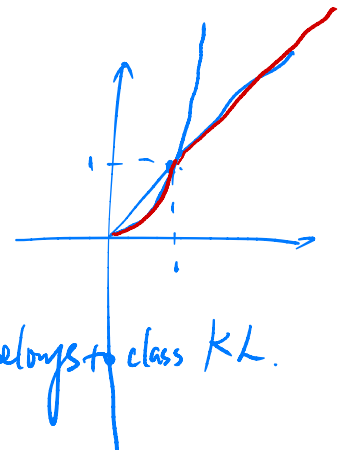
A continuous function $\alpha : [0, a) \rightarrow [0, \infty)$ is said to belong to class $\mathcal{K}$ if it is strictly increasing and $\alpha(0) = 0$. It is of class $\mathcal{K}_\infty$ if $a = \infty$ and $\alpha(r) \rightarrow \infty$ as $r \rightarrow \infty$.

Definition

A continuous function $\beta : [0, a) \times [0, \infty) \rightarrow [0, \infty)$ is said to belong to class $\mathcal{KL}$ if, for each fixed t , the mapping $\beta(r, t)$ belong to class $\mathcal{K}$ with respect to r and, for each fixed r , the mapping $\beta(r, t)$ is decreasing with respect to t and $\beta(r, t) \rightarrow 0$ as $t \rightarrow \infty$.

Examples:

- $\alpha(r) = \arctan(r)$ $r \geq 0$. $\alpha(0) = 0$. $\alpha'(r) = \frac{1}{1+r^2} > 0$.
 $\lim_{r \rightarrow \infty} \alpha(r) = \frac{\pi}{2}$, $\alpha(r)$ belongs to class $\mathcal{K}$, but not class $\mathcal{K}_\infty$.
- $\alpha(r) = r^c$, $c > 0$ belongs to class $\mathcal{K}_\infty$.
- $\alpha(r) = \min(r, r^2)$ belongs to class $\mathcal{K}_\infty$.
- $\beta(r, t) = r^2 e^{-t}$ belongs to class $\mathcal{KL}$.
- $\beta(r, t) = \frac{r}{rt + 1}$ $\beta(r, t) = \frac{1}{t + \frac{1}{r}}$ belongs to class $\mathcal{KL}$.



Lemma (Properties)

Let $\alpha_1, \alpha_2 \in \mathcal{K}$ on $[0, a)$, $\alpha_3, \alpha_4 \in \mathcal{K}_\infty$ and $\beta \in \mathcal{KL}$ and α_i^{-1} denotes the inverse of α_i :

1. α_1^{-1} is defined on $[0, \alpha_1(a))$ and belongs to class $\mathcal{K}$
2. α_3^{-1} is defined on $[0, \infty)$ and belongs to class $\mathcal{K}_\infty$
3. $\alpha_1 \circ \alpha_2$ belongs to class $\mathcal{K}$ $\alpha_1(\alpha_2(r))$
4. $\alpha_3 \circ \alpha_4$ belongs to class $\mathcal{K}_\infty$
5. $\sigma(r, t) = \alpha_1(\beta(\alpha_2(r), t))$ belongs to class $\mathcal{KL}$

Lemma

Let $V : D \rightarrow \mathbb{R}$ be a continuous positive definite function defined on a domain $D \subset \mathbb{R}^n$ that contains the origin. Let $B_r \subset D$ for some $r > 0$. Then, there exist $\alpha_1, \alpha_2 \in \mathcal{K}$ defined on $[0, a]$ such that

$$\alpha_1(\|x\|) \leq V(x) \leq \alpha_2(\|x\|), \quad \forall x \in B_r$$

If $D = \mathbb{R}^n$, α_1 and α_2 are defined on $[0, \infty)$. Moreover, if $V(x)$ is radially unbounded, then $\alpha_1, \alpha_2 \in \mathcal{K}_\infty$.

Example: Consider a quadratic positive definite function

$$V(x) = x^T P x \quad (P \text{ is } PP^T).$$

We know that

$$\lambda_{\min}(P) \|x\|_2^2 \leq x^T P x \leq \lambda_{\max}(P) \|x\|_2^2$$

and so we can define $\alpha_1(r) = \lambda_{\min}(P)r^2$ and $\alpha_2(r) = \lambda_{\max}(P)r^2$

Nonautonomous Systems

Consider the nonautonomous system

$$\dot{x} = f(t, x)$$

Definition

The origin is an **equilibrium point** at $t = 0$ if

$$f(t, 0) = 0, \quad \forall t \geq 0.$$

Definition

The equilibrium point $x = 0$ of the nonautonomous system is

- **Stable** if $\forall \epsilon > 0 \exists \delta = \delta(\epsilon, t_0) > 0 : \|x(t_0)\| < \delta \Rightarrow \|x(t)\| < \epsilon, \quad \forall t \geq t_0 \geq 0$
- **Uniformly Stable (US)** if δ is independent of t_0
- **Asymptotically Stable** if it is stable and $\exists c = c(t_0) : x(t) \rightarrow 0$ as $t \rightarrow \infty$
 $\forall \|x(t_0)\| < c$
- **Uniformly Asymptotically Stable (UAS)** if c is independent of t_0 and the convergence is uniformly in t_0 , that is

$$\forall \eta > 0 \exists T = T(\eta) > 0 : \|x(t)\| < \eta, \quad \forall t \geq t_0 + T(\eta), \forall \|x(t_0)\| < c$$

- **Globally Uniformly Asymptotic Stable (GUAS)** if $\delta(\epsilon)$ can be chosen to satisfies $\lim_{\epsilon \rightarrow \infty} \delta(\epsilon) = \infty$ and

$$\forall (\eta, c) \exists T = T(\eta, c) > 0 : \|x(t)\| < \eta, \quad \forall t \geq t_0 + T(\eta, c), \forall \|x(t_0)\| < c$$

It is easier to define uniform stability and uniform asymptotic stability using comparison functions:

- $x = 0$ is uniformly stable if there exists a class- $\mathcal{K}$ function $\alpha(\cdot)$ and a constant $c > 0$ such that

$$|x(t)| \leq \alpha(|x(t_0)|)$$

for all $t \geq t_0$ and for every initial condition such that $|x(t_0)| \leq c$.

- uniformly asymptotically stable if there exists a class- $\mathcal{KL}$ $\beta(\cdot, \cdot)$ s.t.

$$|x(t)| \leq \beta(|x(t_0)|, t - t_0)$$

for all $t \geq t_0$ and for every initial condition such that $|x(t_0)| \leq c$.

- globally uniformly asymptotically stable if $c = \infty$.
- uniformly exponentially stable if $\beta(r, s) = kre^{-\lambda s}$ for some $k, \lambda > 0$:

$$|x(t)| \leq k|x(t_0)|e^{-\lambda(t-t_0)}$$

for all $t \geq t_0$ and for every initial condition such that $|x(t_0)| \leq c$.

Example:

Consider a nonlinear system

$$\dot{x} = (6t \sin t - 2t)x$$

stable but not uniformly stable.

Its solution is

$$\begin{aligned} x(t) &= x(t_0) \exp \left\{ \int_{t_0}^t (6\tau \sin \tau - 2\tau) d\tau \right\} \\ &= x(t_0) \exp \left\{ \underbrace{6 \sin t - 6t \cos t - t^2}_{\Delta} - \underbrace{6 \sin t_0 + 6t_0 \cos t_0 + t_0^2}_{\delta} \right\} \end{aligned}$$

Note $6 \sin t - 6t \cos t - t^2 \leq 6 + 6t - t^2 = -(t-3)^2 + 15 \leq 15$

We have $\|x(t)\| \leq \|x(t_0)\| \cdot \exp \left\{ \underbrace{15 - 6 \sin t_0 + 6t_0 \cos t_0 + t_0^2}_{\leq \epsilon} \right\} \leq c(t_0) \cdot \|x(t_0)\|$

For any $\epsilon > 0$, choose $\delta(\epsilon, t_0) = \frac{\epsilon}{c(t_0)}$. Then we have

$$\|x(t)\| \leq c(t_0) \cdot \|x(t_0)\| \leq c(t_0) \cdot \frac{\epsilon}{c(t_0)} = \epsilon \Rightarrow \text{stable.}$$

We study the sensitivity of the system solutions due to the changes in t_0 .

Let $t_0 = 2k\pi$. We examine $x(t)$ at $t = t_0 + \pi = (2k+1)\pi$.

$$\begin{aligned} x(t_0 + \pi) &= x(t_0) \cdot \exp \left\{ b \cdot (2k+1)\pi - \left((2k+1)\pi \right)^2 + 12k\pi + (2k\pi)^2 \right\} \\ &= x(t_0) \exp \left\{ (4k+1)(b-\pi)\pi \right\}. \end{aligned}$$

$$\Rightarrow \frac{x(t_0 + \pi)}{x(t_0)} = \exp \left\{ (4k+1)(b-\pi)\pi \right\} \rightarrow \infty \text{ as } k \rightarrow \infty.$$

Therefore, given any $\varepsilon > 0$, there is no $\delta(\varepsilon)$ independent of t_0 that could satisfy the uniform stability definition.

Example: Consider the following system, defined for $t > -1$:

$$\dot{x} = \frac{-x}{1+t}$$

$$\begin{aligned} x(t) &= x(t_0) e^{\int_{t_0}^t \frac{-1}{1+s} ds} = x(t_0) e^{\log(1+s)|_{t_0}^{t_0}} \\ &= x(t_0) e^{\log \frac{1+t_0}{1+t}} = x(t_0) \frac{1+t_0}{1+t} \end{aligned}$$

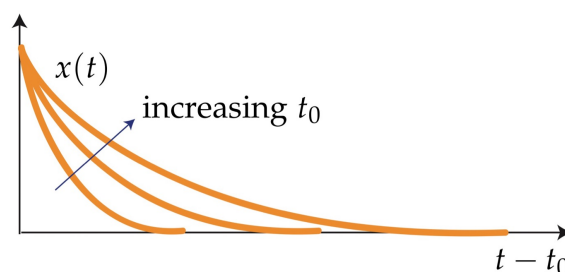
(uniformly stable,
asymptotically stable)

$|x(t)| \leq |x(t_0)| \Rightarrow$ the origin is uniformly stable with $\alpha(r) = r$.

The origin is also asymptotically stable, but not uniformly, because the convergence rate depends on t_0 .

Asy. stable.

$$x(t) = x(t_0) \frac{1+t_0}{1+t_0 + (t-t_0)} = \frac{x(t_0)}{1 + \frac{t-t_0}{1+t_0}}.$$



$$\dot{x} = f(t, x)$$

Time-Dependent Positive Definite Functions

$$V(t, x)$$

Definition:

- $V(t, x)$ is said to be positive semidefinite if $V(t, x) \geq 0$, $V(t, 0) = 0$
- $V(t, x)$ is said to be positive definite if $V(t, x) \geq W_1(x)$, for some positive definite function $W_1(x)$.
- $V(t, x)$ is said to be decreascent if $V(t, x) \leq W_2(x)$, for some positive definite function $W_2(x)$. *渐弱的*
- $V(t, x)$ is radially unbounded if $V(t, x) \rightarrow \infty$, as $|x| \rightarrow \infty$ (W_1 is radially unbounded)

Examples: Let $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$$V(t, x) = (1 + t^2)(x_1^2 + x_2^2)$$

$$W_1(x) = x_1^2 + x_2^2$$

p.d., not decreascent.

$$V(t, x) = (1 + \sin^2 t)(x_1^2 + x_2^2)$$

$$W_1(x) = x_1^2 + x_2^2, W_2(x) = 2(x_1^2 + x_2^2)$$

p.d., decreascent.

$$V(t, x) = \frac{1 + t^2}{2 + t^2}(x_1^2 + x_2^2)$$

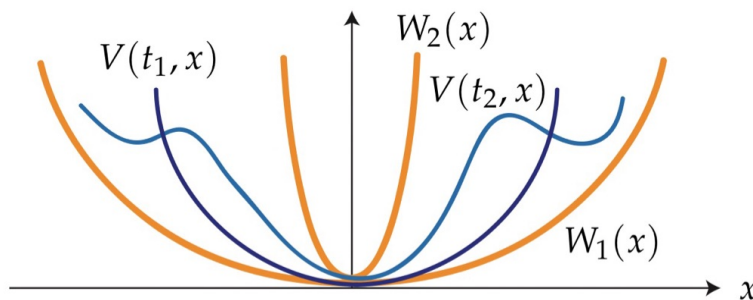
$$W_1(x) = \frac{1}{2}(x_1^2 + x_2^2), W_2(x) = x_1^2 + x_2^2.$$

$$\frac{1}{2} \leq \dots \leq 1$$

Lyapunov's Stability Theorem for Time-Varying Systems

$$\dot{x} = f(t, x)$$

1. If $W_1(x) \leq V(t, x) \leq W_2(x)$ and $\dot{V}(t, x) \triangleq \frac{\partial V}{\partial t} + \frac{\partial V}{\partial x} f(t, x) \leq 0$ for some positive definite functions $W_1(\cdot)$, $W_2(\cdot)$ on a domain D that includes the origin, then $x = 0$ is **uniformly stable**.



2. If, further, $\dot{V}(t, x) \leq -W_3(x) \forall x \in D$ for some positive definite $W_3(\cdot)$, then $x = 0$ is **uniformly asymptotically stable**.
3. If $D = \mathbb{R}^n$ and $W_1(\cdot)$ is radially unbounded, then $x = 0$ is **globally uniformly asymptotically stable**.
4. If $W_i(x) = k_i |x|^a$, $i = 1, 2, 3$, for some constants $k_1, k_2, k_3, a > 0$, then $x = 0$ is **uniformly exponentially stable**.

Example:

Consider the following nonlinear system:

$$\dot{x}_1 = -x_1 - e^{-2t}x_2$$

$$\dot{x}_2 = x_1 - x_2$$

We consider the following Lyapunov function candidate

$$V(t, x) = x_1^2 + \underbrace{c(1 + e^{-2t})}_{0 \leq \leq 1} x_2^2 \quad t: 0 \rightarrow \infty.$$

Since

$$\underbrace{x_1^2 + x_2^2}_{W_1(x) = \|x\|^2} \leq V(t, x) \leq \underbrace{x_1^2 + 2x_2^2}_{W_2(x)} \leq 2(x_1^2 + x_2^2) = 2\|x\|^2$$

$V(t, x)$ is positive definite, decrescent, and radially unbounded.

The derivative of $V(t, x)$ along the system trajectories is given by

$$\dot{V}(t, x) = \frac{\partial V}{\partial t} + \frac{\partial V}{\partial x} \cdot f(t, x).$$

$$= -2e^{-2t}x_2^2 + 2x_1\dot{x}_1 + 2(1 + e^{-2t})x_2\dot{x}_2$$

$$= -2x_1^2 + \underbrace{2x_1x_2 - 2x_2^2}_{(x_1 - x_2)^2} - \underbrace{4e^{-2t}x_2^2}_{4e^{-2t}x_2^2}$$

$$= -(x_1^2 + x_2^2) - (x_1 - x_2)^2 - 4e^{-2t}x_2^2$$

$$\leq -\underbrace{(x_1^2 + x_2^2)}_{W_3(x) = \|x\|^2} \triangleq -W_3(x)$$

Hence, the origin is globally uniformly asymptotically stable.

$$W_1 = W_3 = \|x\|^2, \quad W_2 = 2\|x\|^2 \rightarrow \text{exponentially stable.}$$

Linear Time-Varying Systems

$$\dot{x}(t) = A(t)x(t) \quad (4.2)$$

the solution can be represented as

$$x(t) = \Phi(t, t_0)x(t_0)$$

From linear system theory...

Theorem

The equilibrium point $x = 0$ of the linear system $\dot{x}(t) = A(t)x(t)$ is GUAS if and only if

$$\|\Phi(t, t_0)\| \leq ke^{-\lambda(t-t_0)}, \quad \forall t \geq t_0 \geq 0$$

for some positive constants k and λ .

Remark that

1. for linear systems GUAS $\Leftrightarrow$ Exponential stability
2. for linear time-varying system, GUAS cannot be characterized by the location of the eigenvalues of A .

Example

$$A(t) = \begin{bmatrix} -1 + 1.5\cos^2 t & 1 - 1.5\sin t \cos t \\ 1 - 1.5\sin t \cos t & -1 + 1.5\sin^2 t \end{bmatrix}$$

For each t , $\lambda_i(A(t)) = -0.25 \pm 0.25\sqrt{7}j$. Yet, the origin is unstable. $\Phi(t, 0)$ is given by

$$\Phi(t, 0) = \begin{bmatrix} e^{0.5t} \cos t & e^{-t} \sin t \\ -e^{0.5t} \sin t & e^{-t} \cos t \end{bmatrix}$$

Corollary: The system (4.2) is uniformly asymptotically stable if there exists $\lambda > 0$ such that

$$\lambda_i(A(t) + A^T(t)) \leq -\lambda, \quad \forall t \geq 0, \quad i = 1, \dots, n.$$

Proof: Let $V(t, x) = x^T x$.

$$\text{Then } \dot{V} = \dot{x}^T x + x^T \dot{x}$$

$$= (A(t)x)^T x + x^T A(t)x$$

$$= x^T (A^T(t) + A(t)) x \leq -\lambda x^T x = -\lambda V.$$

Corollary : Assume that there exists $P(t) \in \mathbb{R}^{n \times n}$ which is continuously differentiable and symmetric, and there exist $0 < c_1 < c_2 < \infty$ such that

$$c_1 I \leq P(t) \leq c_2 I, \quad \forall t \geq 0$$

Further assume for some $Q(t) \in \mathbb{R}^{n \times n}$, continuous and symmetric such that

$$Q(t) \geq c_3 I > 0, \quad \forall t \geq 0$$

$$\dot{P}(t) + P(t)A(t) + A^T(t)P(t) = -Q(t)$$

Then the system (4.2) is globally uniformly asymptotically stable.

Proof: Let $V(t, x) = x^T P(t) x \Rightarrow c_1 \overline{w_1(x)} \leq V(t, x) \leq c_2 \overline{w_2(x)}$. *pd. decreasing.*

$$\dot{V} = \dot{x}^T P(t) x + x^T \dot{P}(t) x + x^T P(t) \dot{x}$$

$$= -x^T Q x \leq -c_3 \overline{w_3(x)}$$

Corollary : Assume that at any time $t \geq 0$, the eigenvalues of $A(t)$ all have

negative real parts. In addition, if $A(t)$ is bounded, and $\int_0^\infty A^T(\tau) A(\tau) d\tau < \infty$,

then (4.2) is globally uniformly asymptotically stable.

$$\begin{aligned} \dot{x} &= f(x) \\ \dot{x} &= \frac{\partial f}{\partial x} \Big|_{x=0} \cdot x \\ &\downarrow \\ \dot{x} &= A x \end{aligned}$$

$$\begin{aligned} \dot{x} &= f(t, x) \\ &\downarrow \text{linearization} \\ \dot{x} &= \frac{\partial f(t, x)}{\partial x} \Big|_{x=0} \cdot x \\ &\downarrow \\ \dot{x} &= A(t) x \end{aligned}$$

Theorem

Let $x = 0$ be an equilibrium point for the nonlinear system

$$\dot{x} = f(t, x) \quad (*)$$

where $f : [0, \infty) \times D \rightarrow \mathbb{R}^n$ is C^1 , $D = \{x \in \mathbb{R}^2 : \|x\|_2 < r\}$, and the Jacobian matrix $\frac{\partial f}{\partial x}$ is bounded and Lipschitz on D , uniformly in t . Let

$$A(t) = \frac{\partial f}{\partial x}(t, x) \Big|_{x=0}$$

Then, the origin is an exponentially stable equilibrium point of (*) if and only if it is an exponentially stable equilibrium point for the linear system $\dot{x} = A(t)x$.

Barbalat's Lemma

How about the asymptotical stability when $\dot{V}(t, x)$ is only negative semi-definite? For autonomous systems, LaSalle's invariance theorem can be used. But it is not valid for nonautonomous systems.

Barbalat's Lemma is a purely mathematical result concerning the asymptotical properties of functions and their derivatives.

$$f(t) \text{ converges} \not\Rightarrow \dot{f} \rightarrow 0 \quad \text{No!}$$

$$f(t) = e^{-t} \sin(e^{2t}) \quad \dot{f}(t) = -e^{-t} \sin(e^{2t}) + \underline{e^{-t} \cdot \cos(e^{2t}) \cdot 2e^{2t}}$$

$$\dot{f} \rightarrow 0 \not\Rightarrow f(t) \text{ converges.} \quad \text{No!}$$

$$f(t) = \ln t \quad \dot{f}(t) = \frac{1}{t}$$

$\chi^{\frac{3}{2}}$

Definition: Uniform Continuity (一致连续)

A function $f: \mathbb{R} \rightarrow \mathbb{R}$ is said to be uniformly continuous if

$$\forall \varepsilon > 0, \exists \delta(\varepsilon) > 0, \forall |t_2 - t_1| \leq \delta \implies |f(t_2) - f(t_1)| \leq \varepsilon$$

Lemma (Barbalat):

If the differentiable function $f(t)$ ^① has a finite limit as $t \rightarrow \infty$, and if $\dot{f}(t)$ ^② is uniformly continuous, then $\lim_{t \rightarrow \infty} \dot{f}(t) = 0$. (② is bounded)

Note: A sufficient condition for a differentiable function to be uniformly continuous is that its derivative be bounded.

Proof: Assume that $\dot{f}(t)$ does not approach zero as $t \rightarrow \infty$. Then $\exists \varepsilon_0 > 0, \forall T > 0, \exists t > T, |\dot{f}(t)| \geq \varepsilon_0$. Therefore, we can get an infinite sequence of t_i 's (such that $t_i \rightarrow \infty$ as $i \rightarrow \infty$) such that $|\dot{f}(t_i)| \geq \varepsilon_0$.

Since $\dot{f}(t)$ is assumed to be uniformly continuous, $\exists \eta > 0$, such that $|\dot{f}(t') - \dot{f}(t'')| < \frac{\varepsilon_0}{2}, \forall |t' - t''| < \eta$.

$$\forall |t-t_i| < \eta, \quad |\dot{f}(t)| - |\dot{f}(t_i)| \leq |\dot{f}(t) - \dot{f}(t_i)| < \frac{\epsilon_0}{2}.$$

$$\text{for all } t_i, \quad \frac{\epsilon_0}{2} \leq |\dot{f}(t_i)| - \frac{\epsilon_0}{2} \leq |\dot{f}(t)| \leq |\dot{f}(t_i)| + \frac{\epsilon_0}{2}$$

$$\left| \int_{t_i-\eta}^{t_i+\eta} \dot{f}(t) dt \right| = \int_{t_i-\eta}^{t_i+\eta} |\dot{f}(t)| dt \geq 2\eta \cdot \frac{\epsilon_0}{2} = \epsilon_0 \eta.$$

$$= |f(t_i+\eta) - f(t_i-\eta)|.$$

This result would contradict with the fact $f(t)$ has a finite limit.

Lemma ("Lyapunov-Like"):

If a scalar function $V(t, x)$ satisfies the following conditions:

- $V(t, x)$ is lower bounded
 - $\dot{V}(t, x)$ is negative semidefinite $(\dot{V} \leq 0)$
 - $\dot{V}(t, x)$ is uniformly continuous $(\dot{V} \text{ bounded})$
- $\Rightarrow V(t, x)$ has a finite limit.

then $\lim_{t \rightarrow \infty} \dot{V}(t, x) = 0$.

Example:

In adaptive control, we will often encounter the following nonautonomous system

$$\dot{e} = -e + \varphi(t)\tilde{\theta}$$

$$\dot{\tilde{\theta}} = -\varphi(t)e$$

where $\varphi(t)$ is a bounded continuous function, e and $\tilde{\theta}$ are the two states of the closed-loop system, representing the tracking error and parameter error.

Consider the quadratic scalar function

$$V = \frac{1}{2}e^2 + \frac{1}{2}\tilde{\theta}^2$$

Its derivative is

$$\dot{V} = e \cdot \dot{e} + \tilde{\theta} \dot{\tilde{\theta}}$$

$$= -e^2 + \varphi(t)e\tilde{\theta} - \tilde{\theta}\varphi(t)e.$$

$$= -e^2 \leq 0 \quad (\text{NSD}).$$

Check $\ddot{V} = -2e \cdot \dot{e} = -2e(-e + \varphi(t)\tilde{\theta})$. (16)

Note that $\dot{V} \leq 0$, which implies that $V(t) \leq V(0)$. Therefore, e , $\tilde{\theta}$ are bounded.

With $\varphi(t)$ being bounded, we can get $\ddot{V}$ is also bounded. From Barbalat's lemma, we can conclude that $\lim_{t \rightarrow \infty} \dot{V}(t) = 0$, i.e., $\lim_{t \rightarrow \infty} e(t) = 0$.

Theorem (LaSalle-Yoshizawa):

Let $x=0$ be an equilibrium point of $\dot{x} = f(t, x)$ and suppose that $f(t, x)$ is locally Lipschitz in x and uniformly in t . Let $V(t, x)$ be a continuously differentiable function such that

$$\alpha_1(|x|) \leq V(t, x) \leq \alpha_2(|x|)$$

$$\dot{V} = \frac{\partial V}{\partial t} + \frac{\partial V}{\partial x} f(t, x) \leq -W(x) \leq 0$$

$\forall t \geq 0, x \in \mathbb{R}^n$, where $\alpha_1(\cdot)$ and $\alpha_2(\cdot)$ are class $\mathcal{K}_\infty$ functions and $W(x)$ is a continuous function. Then all solutions of $\dot{x} = f(t, x)$ satisfy

$$\lim_{t \rightarrow \infty} W(x(t)) = 0.$$

In addition, if $W(x)$ is positive definite, $x=0$ is globally uniformly asymptotically stable.

Proof: Revisit: f has a finite limit, f uniformly continuous $\rightarrow \dot{f} \rightarrow 0$
 $\int_0^t W(x(\tau)) d\tau$ has a finite limit, $W(x(t))$ is uniformly continuous $\rightarrow W(x(t)) \rightarrow 0$.

Note $W(x) \geq 0$, $\int_0^t W(x(\tau)) d\tau \uparrow$.

$$\left. \begin{aligned} \text{From } \dot{V} \leq -W(x), \quad V(t) - V(0) &\leq -\int_0^t W(x(\tau)) d\tau \\ \Rightarrow \int_0^t W(x(\tau)) d\tau &\leq V(0) - V(t) \leq V(0) \end{aligned} \right\} \Rightarrow \int_0^t W(x(\tau)) d\tau \text{ has a finite limit.}$$

Since $\dot{V} \leq 0$, $\alpha_1(\|x(t)\|) \leq V(t, x) \leq \alpha_2(\|x(t)\|) \Rightarrow V$ is bounded. $\|x(t)\| \leq \alpha_1^{-1}(V(t, x))$ is bounded.

Assume that $\|x(t)\| \leq R$ ($\cong \alpha_1^{-1}(V(0))$). Note that a continuous function is uniformly continuous on a closed set. $W(x)$ is uniformly continuous in x .

$$\forall t_2 - t_1 > 0.$$

$$\begin{aligned} |x(t_2) - x(t_1)| &= \left| \int_{t_1}^{t_2} f(\tau, x(\tau)) d\tau \right| \stackrel{\text{locally Lipschitz}}{\leq} L_R \int_{t_1}^{t_2} \|x(\tau)\| d\tau \\ &\leq L_R \cdot R |t_2 - t_1|. \quad (\|x\| \in R) \end{aligned}$$

Then $x(t)$ is uniformly continuous in t , and thus $W(x(t))$ is also uniformly continuous in t .

From Barbalat's Lemma, $\lim_{t \rightarrow \infty} W(x(t)) = 0$.

Boundedness and Ultimate Boundedness

The concepts of stability in the sense of Lyapunov are formulated with respect to an equilibrium point. Often, systems are designed to operate in the presence of disturbances and other uncertainties.

Until now we have used Lyapunov theory to study the behavior of the system about the equilibrium point.

What happens when the system does not have any equilibrium point?

We will see that Lyapunov analysis can be used to show boundedness of the solution of the state equation.

Example

$$\begin{aligned} \dot{x} &= -kx + \delta \sin t \quad \rightarrow \quad \frac{\delta}{R} \\ \dot{x} &= \underline{-x} + \delta \sin t, \quad x(t_0) = a, \quad a > \delta > 0 \end{aligned}$$

There are no equilibrium points!

Its solution is given by

$$x(t) = e^{-(t-t_0)} x(t_0) + \delta \int_{t_0}^t e^{-(t-\tau)} \sin \tau d\tau.$$

The solution satisfies

$$\begin{aligned} \|x(t)\| &\leq e^{-(t-t_0)} \|x(t_0)\| + \delta \int_{t_0}^t e^{-(t-\tau)} d\tau \\ &= e^{-(t-t_0)} a + \delta (1 - e^{-(t-t_0)}) \\ &= \delta + (a - \delta) e^{-(t-t_0)} \leq \delta + a - \delta = a. \end{aligned}$$

$$\begin{aligned} \dot{x} + x &= \delta \sin t \\ e^t \dot{x} + e^t x &= \delta e^t \sin t \\ \frac{d}{dt} (e^t x) &= \delta e^t \sin t \end{aligned}$$

$$\left(e^{-t} \cdot \int_{t_0}^t e^{\tau} d\tau \right)$$

which shows that the solution is uniformly bounded. Further, for any constant $b \in (\delta, a)$, it can be easily seen that

$$\|x(t)\| \leq b, \quad \forall t \geq t_0 + \ln \frac{a-\delta}{b-\delta}$$

$$\begin{aligned} b &= \delta + (a-\delta)e^{-(t-t_0)} \\ e^{-(t-t_0)} &= \frac{b-\delta}{a-\delta} \\ -t+t_0 &= \ln \frac{b-\delta}{a-\delta} \\ \Rightarrow t &= t_0 - \ln \frac{b-\delta}{a-\delta} \end{aligned}$$

The bound b , which again is independent of t_0 , gives a better estimate of the solution after a transient period has passed. In this case, the solution is said to be **uniformly ultimately bounded (UUB)** and b is called the **ultimate bound**.

This can be also done via **Lyapunov analysis** without the explicit solution of the state equation. Consider $V = \frac{1}{2}x^2$. Its derivative is

$$\begin{aligned} \dot{V} &= x \cdot \dot{x} = x(-x + \delta \sin t) = -x^2 + \delta x \sin t \\ &\leq -x^2 + \delta |x| = -|x|(|x| - \delta). \end{aligned}$$

$\dot{V}$ is negative definite outside the set $\{|x| \leq \delta\}$.

With $c > \frac{1}{2}\delta^2$, ($|x| > \delta$), solution starting in the set $\{V(x) \leq c\}$ will remain there for all future time. Hence, the solutions are uniformly bounded [↓] invariant set.

Pick up any number ε such that $\frac{\delta^2}{2} < \varepsilon < c$. Then $\dot{V} < 0$ in the set $\{\varepsilon \leq V \leq c\}$ which shows that, in this set, V will decrease monotonically until the solution enters the set $\{V(t) \leq \varepsilon\}$. From that time on, the solution cannot leave the set $\{V \leq \varepsilon\}$ since $\dot{V} < 0$ on the boundary of $V = \varepsilon$. Thus, we can conclude that the solution is uniformly bounded with the ultimate bound $|x| \leq \sqrt{2\varepsilon}$.

Definition

The solutions of $\dot{x} = f(t, x)$ are

- **Uniformly Bounded (UB)** if there exists a $c > 0$, independent of $t_0 \geq 0$ such that

$$\forall a \in (0, c) \exists \beta = \beta(a) > 0 : \|x(t_0)\| \leq a \Rightarrow \|x(t)\| \leq \beta, \forall t \geq t_0$$

- **Globally Uniformly Bounded (GUB)** if a can be arbitrarily large
- **Uniformly Ultimately Bounded (UUB)** with ultimate bound b , if there exists $b > 0, c > 0$ (independent of t_0) such that

$$\forall a \in (0, c) \exists T = T(a, b) \geq 0 : \|x(t_0)\| \leq a \Rightarrow \|x(t)\| \leq b, \forall t \geq t_0 + T,$$

- **Globally Uniformly Ultimately Bounded (GUUB)** if a can be arbitrarily large.

Remark: For a Lyapunov function candidate V .

$$\boxed{\dot{V} \leq -\beta V + c,} \quad \beta, c > 0. \Rightarrow \text{UUB}$$

$$\dot{V} + \beta V \leq c.$$

$$e^{\beta t} \dot{V} + \beta e^{\beta t} V \leq c e^{\beta t}$$

$$\frac{d}{dt} (e^{\beta t} V(t)) \leq c e^{\beta t}.$$

$$e^{\beta t} V(t) - e^{\beta t_0} V(t_0) \leq c \cdot \int_{t_0}^t e^{\beta \tau} d\tau = \frac{c}{\beta} (e^{\beta t} - e^{\beta t_0})$$

$$\Rightarrow V(t) \leq e^{-\beta(t-t_0)} V(t_0) + \frac{c}{\beta} (1 - e^{-\beta(t-t_0)})$$

Input-to-state stability (ISS).

For a linear time-invariant system

$$\dot{x} = Ax + Bu$$

with A Hurwitz. The solution is given by

$$x(t) = e^{A(t-t_0)}x(t_0) + \int_{t_0}^t e^{A(t-\tau)}Bu(\tau)d\tau$$

Using the bound

$$\|e^{A(t-t_0)}\| \leq ke^{-\lambda(t-t_0)}$$

we conclude that

$$\begin{aligned}\|x(t)\| &\leq ke^{-\lambda(t-t_0)}\|x(t_0)\| + \int_{t_0}^t ke^{-\lambda(t-\tau)}\|B\|\|u(\tau)\|d\tau \\ &\leq ke^{-\lambda(t-t_0)}\|x(t_0)\| + \frac{k}{\lambda}(1 - e^{-\lambda(t-t_0)})\|B\| \sup_{t_0 \leq \tau \leq t} \|u(\tau)\| \\ &\leq ke^{-\lambda(t-t_0)}\|x(t_0)\| + \frac{k}{\lambda}\|B\| \sup_{t_0 \leq \tau \leq t} \|u(\tau)\|\end{aligned}$$

Does this hold for a general nonlinear system?

$$\dot{x} = f(t, x, u)$$

And in what conditions? Is it sufficient to have GUAS of the unforced system? NO!

Example

$$\dot{x} = -3x + (1 + 2x^2)u$$

when $u = 0$, the equilibrium point $x = 0$ is GAS. Yet, when $x(0) = 2$ and $u(t) = 1$, the solution is given by

$$x(t) = \frac{3 - e^t}{3 - 2e^t}$$

which is unbounded. It even has a finite escape time.

Definition

The system

(输入到状态稳定) $\dot{x} = f(t, x, u)$

is said to be input-to-state stable (ISS) if there exist $\beta \in \mathcal{KL}$, and $\gamma \in \mathcal{K}$ such that for any initial state $x(t_0)$ and any bounded input $u(t)$, the solution $x(t)$ exists for all $t \geq t_0$ and satisfies

$$\|x(t)\| \leq \beta(\|x(t_0)\|, t - t_0) + \gamma\left(\sup_{t_0 \leq \tau \leq t} \|u(\tau)\|\right)$$

Remarks:

1. For any bounded input $u(t)$, the state $x(t)$ will be bounded
2. As t increases, the state $x(t)$ will be ultimately bounded by a class $\mathcal{K}$ function of $\sup_{t \geq t_0} \|u(t)\|$
3. If $u(t) \rightarrow 0$ as $t \rightarrow \infty$ then $x(t) \rightarrow 0$ as $t \rightarrow \infty$
4. Since for $u(t) = 0$, $\|x(t)\| \leq \beta(\|x(t_0)\|, t - t_0) \Leftrightarrow x = 0$ of the unforced system is GUAS.

Theorem

Let $V : [0, \infty) \times \mathbb{R}^n \rightarrow \mathbb{R}$ be a C^1 function such that

$$\alpha_1(\|x\|) \leq V(t, x) \leq \alpha_2(\|x\|)$$

$$\frac{\partial V}{\partial t} + \frac{\partial V}{\partial x} f(t, x, u) \leq -W_3(x), \quad \forall \|x\| \geq \rho(u) > 0$$

$\forall (t, x, u) \in [0, \infty) \times \mathbb{R}^n \times \mathbb{R}^n$, $\alpha_1, \alpha_2 \in \mathcal{K}_\infty$, $\rho \in \mathcal{K}$, $W_3(x) > 0$ and C^0 . Then the system $\dot{x} = f(t, x, u)$ is ISS with $\gamma = \alpha_1^{-1} \circ \alpha_2 \circ \rho$.

Lemma

$$\dot{x} = f(t, x, u)$$

Suppose that $f(t, x, u)$ is C^1 and globally Lipschitz in (x, u) , uniformly in t . If the unforced system has a globally exponentially stable equilibrium point at $x = 0$, then the system $\dot{x} = f(t, x, u)$ is ISS.

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = \tau.$$

$$\tau = -k_p q - k_v \dot{q}$$

$$s = \dot{q} + \lambda q, \quad \lambda > 0.$$

$$M\dot{s} + \dots = (-\partial \eta \xi(\lambda)).$$

$$\Rightarrow \dot{q} = -\lambda q + s$$

$\downarrow$ State
 $\downarrow$ input
 (ISS)

s bounded $\Rightarrow q, \dot{q}$ bounded.
 $s \rightarrow 0 \Rightarrow q \rightarrow 0, \dot{q} \rightarrow 0.$

Cascade System

Consider the following system

$$\dot{x}_1 = f_1(t, x_1, x_2) \quad (*)$$

where the equilibrium point $x_1 = 0$ of $\dot{x}_1 = f_1(t, x_1, 0)$, is GUAS in cascade with

$$\dot{x}_2 = f_2(t, x_2),$$

where the equilibrium point $x_2 = 0$ of $\dot{x}_2 = f_2(t, x_2)$, is GUAS.

Under what conditions will the origin $x = (x_1, x_2) = 0$ of the cascade system be GUAS?

Lemma

If system (*) is ISS with x_2 as input, then the origin of the cascade system is GUAS.